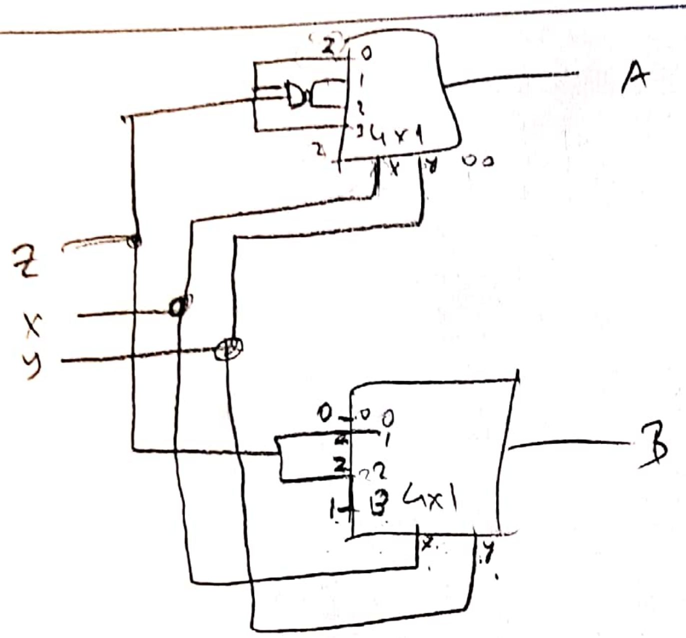
①



x	y	z	A	B
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

2 NAND 2

||

A	B	NAND
0	0	1
0	1	1
1	0	1
1	1	0

b) Show functionality.

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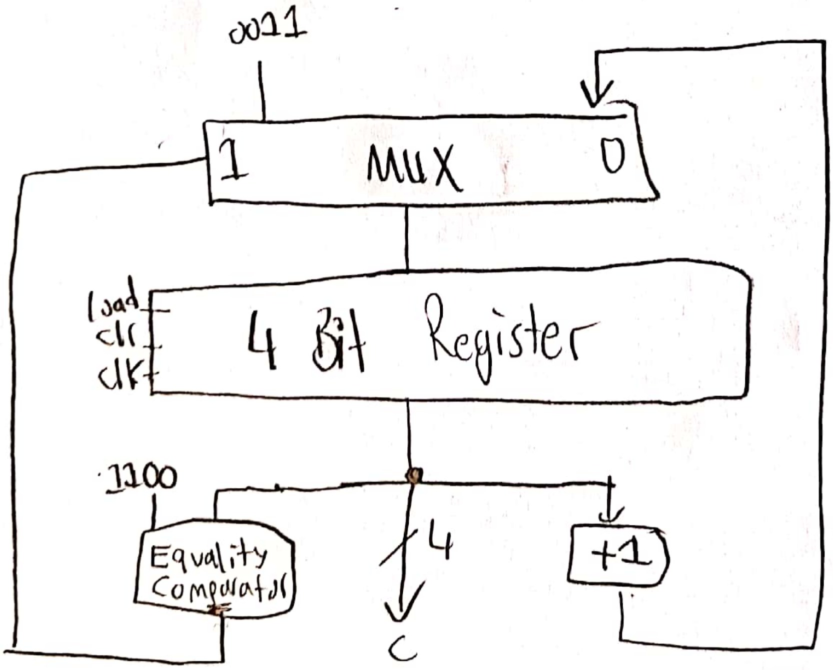
② Design counter that counts in sequence: 3,4,5,6,7,8,9,10,11,12,3



Bilgi: Gelen sayı 12'ye eşit ise tekrar 3'e dönmesi için mux 1. kısmına bağla. Değilse mesela 4 geldiyse +1 artır 0.'ya bağ
Ortadaki C ise o ankini çıkart.

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③ Implement 4 bit register → 0 current, 1 loaded, 4 condition → 0→load 1→current 2→left 3→right.
slaytta uygması var.

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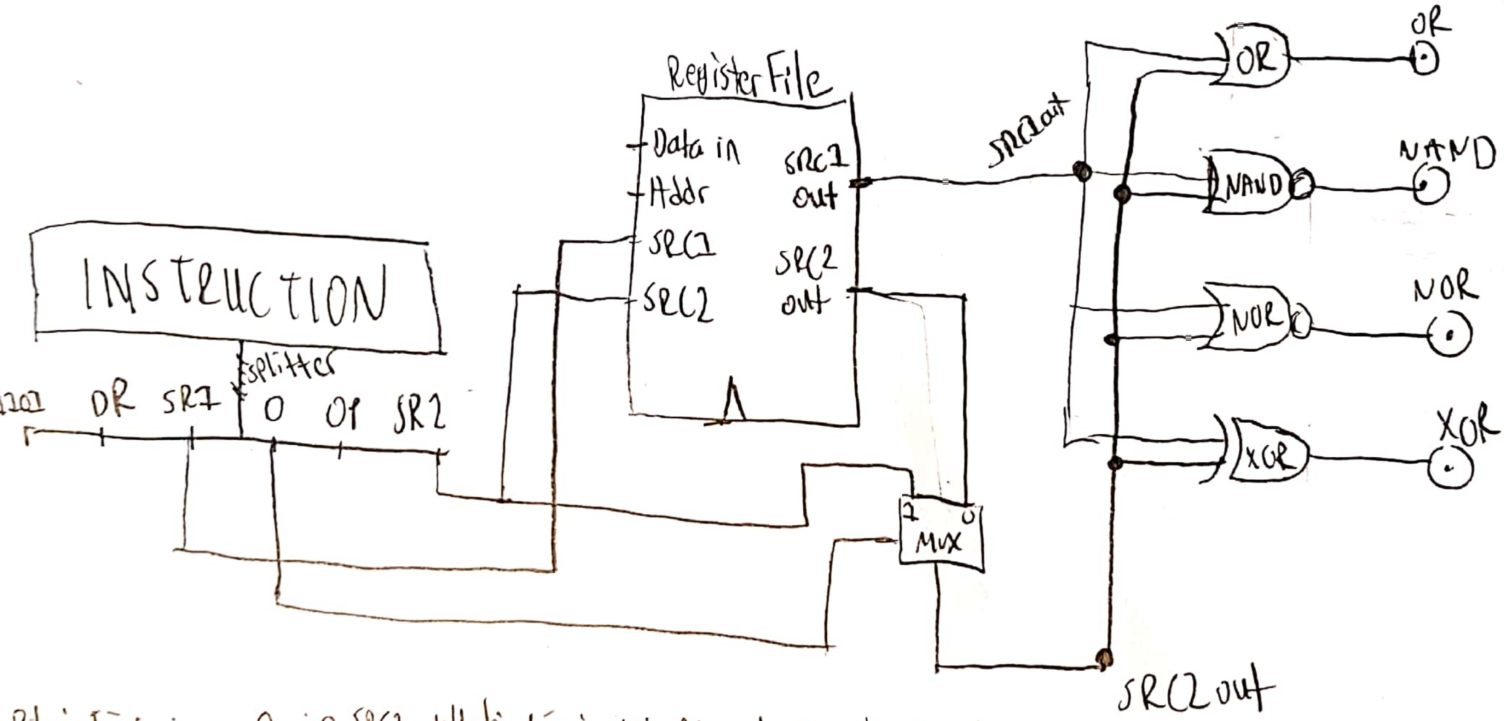
④ Use opcode 1101 imp. four bitwise OR, NAND, NOR, XOR. 16 bit ins.

1101	DR	SR1	0	OP	SR2

, IR[5] → whether use imm val of register for the source value IR[4:3] → four logical op. to be performed.
Design the circuit.



Bilgi: Eğer imm=0 ise SRC2 outtaki değeri yani SR2 adresinin içindeki değeri; imm1 ise direkt S2 adresini instructio
dan al. → Bunun için mux kullandık.

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⑤ RTL to find mean of the number. find mean X.
If X → even: output: X/2 . If X → odd → round to $\lceil X/2 \rceil$
ex: X=10, output = 5,
X=11, output: 6
Design RTL, circuit, fsm